

GridCast: Probabilistic Grid Stress Forecasting for Dynamic Data Center Power Allocation

Project Proposal | Draft for Review

[Author names omitted for review]

Abstract

Data centers increasingly strain power grid infrastructure, yet flexible power contracts remain underutilised because utilities cannot forecast grid stress with enough confidence to underwrite contractual commitments. GRIDCAST is a probabilistic deep-learning pipeline that combines GRAPHCAST and GENCAST (Google DeepMind’s deterministic and ensemble weather models) with regional grid context from the EIA API and node-level stress signals from Grid Status locational marginal price (LMP) data. Together, these three layers produce confidence-bounded grid stress estimates at the transmission-node level, giving utilities the uncertainty-aware signal needed to offer dynamic power contracts. A key advantage is GRAPHCAST’s trainability: unlike physics-locked NWP models, it can be fine-tuned on recent observational data to adapt to shifting climate trends.

1 Introduction and Problem Statement

The power grid is overbuilt by design, and most excess capacity sits idle nearly every day. Under flexible power contracts, data centers would commit to drawing less during grid stress events—compensating through workload shifting or graceful degradation—in exchange for access to that idle capacity at preferential rates.

The bottleneck is not a lack of forecasting. Utilities already use meteorological inputs for demand forecasting. The problem is the *absence of forecasting confident enough to underwrite contractual commitments*. Grid stress emerges from compounding tail-risk scenarios: a heatwave extending longer than forecast, a cold snap of unexpected severity, or high demand coinciding with renewable underperformance. Deterministic point forecasts cannot communicate the probability of such scenarios. Without uncertainty bounds, no utility can confidently offer a tiered contract, and no data center can reliably plan around one.

GRIDCAST addresses this by producing confidence-bounded, node-level stress estimates over a 10-day horizon.

2 Why AI Weather Forecasting

GRAPHCAST operates at 0.25-degree global resolution and can be queried over precise coordinates, enabling node-level reasoning rather than regional averaging. As a learned model trained on the ERA5 reanalysis archive, it can be fine-tuned on recent data to adapt to shifting climate patterns, which physics-locked NWP systems cannot do. It also produces 10-day global forecasts in under one minute, enabling near-real-time updates for hourly contract triggers.

GENCAST extends GRAPHCAST by generating an *ensemble* of plausible futures rather than a single deterministic trajectory. A utility does not need to know a heatwave *will* occur; it needs to

know the probability and severity range. This ensemble output is what makes the downstream stress signal actionable for contract design.

3 Proposed Approach

3.1 Three-Layer Pipeline

Layer 1: Probabilistic Weather Forecasting (GRAPHCAST + GENCAST). GRAPHCAST forecasts key atmospheric variables at 6-hour intervals over a 10-day horizon. ERA5 reanalysis serves as the training ground truth; at inference time, GRAPHCAST forecasts replace ERA5. GENCAST wraps GRAPHCAST’s output with a diffusion-based ensemble, producing a distribution over plausible futures with explicit confidence intervals that propagate through the downstream layers.

Layer 2: Regional Grid Context (EIA API). EIA hourly demand and fuel-mix data from Form EIA-930 (`/v2/electricity/rto/`) are used to train the weather-to-load relationship for each Balancing Authority. EIA data also captures supply-side context: renewable penetration, available headroom, and the forecast-versus-actual demand delta. These variables are essential for identifying compounding scenarios where demand spikes coincide with renewable underperformance.

Layer 3: Node-Level Stress Signal (Grid Status LMP). Locational Marginal Price at a target node is the wholesale market’s real-time expression of congestion and supply tightness at that specific location, providing the node-level resolution that EIA cannot supply. Historical nodal LMP from the open-source `gridstatus` Python library is used to train a mapping from regional stress to node-level price behaviour; day-ahead LMP serves as a forward signal at inference time.

3.2 Key Input Variables

Table 1 summarises the input features, their sources, and their roles in the pipeline.

Table 1. Key input variables, data sources, and pipeline roles.

Category	Variable	Source	Role
Weather	2 m Temperature	GRAPHCAST/ ERA5	Primary demand driver
Weather	10 m Wind Speed	GRAPHCAST/ ERA5	Renewable generation proxy
Weather	Ensemble Spread	GENCAST	Uncertainty quantification
Grid (EIA)	Hourly Actual Demand	EIA <code>/rto/region-data</code>	Training target variable
Grid (EIA)	Generation by Fuel Type	EIA <code>/rto/fuel-type</code>	Renewable underperformance
Grid (LMP)	Real-Time Nodal LMP	Grid Status	Node-level stress signal
Grid (LMP)	LMP Congestion Component	Grid Status	Transmission constraint
Engineered	Hour / Day / Month	Derived	Demand periodicity
Engineered	Demand / Capacity Ratio	Derived from EIA	Interpretable stress metric

3.3 Training and Output

The weather-to-grid-stress mapping is trained on ERA5 paired with EIA historical data from 2018 to present, the period over which both are available at compatible resolution. Training on reanalysis rather than forecast outputs is essential: forecasts carry errors that would otherwise propagate into the learned mapping.

At inference time, GRAPHCAST and GENCAST replace ERA5, and the pipeline produces a confidence-bounded probability distribution over grid stress at a target node over the 10-day horizon. This is operationalised as a *recommended power allocation percentage*: the share of excess capacity at a given node that can safely be committed to a data center operator, conditioned on forecast confidence.

4 Validation

The pipeline is validated against historical EIA load data and nodal LMP during documented stress events. Three well-documented cases are used, covering distinct failure modes and grid topologies:

- **2021 Texas Winter Storm.** Cold-driven demand surge; extreme exceedance of forecast demand in ERCOT territory.
- **2022 Pacific Northwest Heat Dome.** Sustained multi-day heat event testing the model’s ability to maintain calibration over a prolonged horizon.
- **2023 PJM Summer Stress Events.** Compounding of high load with renewable underperformance across a large, highly interconnected grid topology.

5 Value Proposition

This project demonstrates that a probabilistic, node-level forecasting layer for dynamic power contracts is technically viable today using existing open-source tools and public data.

- **Utility Providers.** Node-level probabilistic forecasts let utilities offer tiered contracts with confidence, converting idle excess capacity into revenue.
- **Grid Operators (PJM, CAISO, MISO).** Better advance forecasting reduces costly last-minute interventions and improves visibility into renewable supply fluctuations.
- **Data Centers.** Forecast-driven contracts unlock excess grid capacity, reducing operational costs and enabling more sustainable scaling of AI compute.
- **Hardware Providers (e.g., NVIDIA).** Power availability is a primary bottleneck to data center expansion. Unlocking excess grid capacity has clear strategic value for businesses whose growth depends on compute scaling.

A Data Sources

A.1 EIA API v2: Hourly Electric Grid Monitor

Source: U.S. Energy Information Administration, Form EIA-930. Free public API at api.eia.gov/v2/electricity/rto/ (free key registration at eia.gov/opendata). Data are organised by Balancing Authority at hourly resolution, with historical depth from July 2015 for demand and generation totals and July 2018 for generation by fuel type.

Listing 1. Example EIA API query for PJM hourly demand.

```
GET https://api.eia.gov/v2/electricity/rto/region-data/data/
?frequency=hourly
&data[]=value
&facets[respondent][]=PJM
```

```
&facets[type][]=D
&sort[0][column]=period
&api_key=YOUR_KEY
```

Table 2. EIA /rto/region-data sample — PJM, 27 July 2023.

Period	Resp.	Type	Type Name	Value	Units
2023-07-27T17	PJM	D	Demand	152,340	MWh
2023-07-27T17	PJM	DF	Forecast Demand	148,900	MWh
2023-07-27T17	PJM	NG	Net Generation	149,820	MWh
2023-07-27T17	PJM	TI	Total Interchange	−2,520	MWh

The forecast-versus-actual delta (152,340 vs. 148,900 MWh) on a peak summer afternoon illustrates the surprise stress deviation the pipeline is designed to anticipate *before* it occurs.

Table 3. EIA /rto/fuel-type-data sample — CAISO, 27 July 2023.

Period	Resp.	Fuel	Fuel Name	Value	Units
2023-07-27T15	CISO	SUN	Solar	9,842	MWh
2023-07-27T15	CISO	NG	Natural Gas	14,560	MWh
2023-07-27T18	CISO	SUN	Solar	3,410	MWh
2023-07-27T18	CISO	NG	Natural Gas	18,940	MWh

The drop in solar and corresponding rise in gas dispatch between 15:00 and 18:00 illustrates the afternoon duck curve—a daily structural stress window in CAISO territory.

A.2 Grid Status: Nodal LMP Data

Source: gridstatus.io. Open-source Python library (`pip install gridstatus`) for recent data; commercial API tier for full historical depth. Covers PJM, CAISO, MISO, ERCOT, ISO-NE, NYISO, and SPP at 5-minute real-time and hourly day-ahead resolution. LMP decomposes into energy, congestion, and loss components. The congestion component isolates transmission bottlenecks and is the primary node-level stress signal used in GRIDCAST.

Listing 2. Fetching PJM real-time LMP via `gridstatus`.

```
import gridstatus

pjm = gridstatus.PJM()
df = pjm.get_lmp(
    date="2023-07-27",
    market="REAL_TIME_5_MIN",
    locations="ALL"
)
```

Table 4. Grid Status real-time LMP — PJM, 27 July 2023 (peak hour). All prices in \$/MWh.

Time	Location	LMP	Energy	Congestion	Loss
2023-07-27 17:00	Dominion Hub	142.30	98.40	38.70	5.20
2023-07-27 17:00	AEP-Dayton Hub	101.15	98.40	1.40	1.35
2023-07-27 17:00	Western Hub	99.80	98.40	0.20	1.20
2023-07-27 17:00	N. Illinois Hub	103.60	98.40	3.70	1.50

Dominion Hub (Northern Virginia, a major data center corridor) shows a congestion component of \$38.70/MWh—nearly 40% of total LMP—while other hubs show near-zero congestion at the same system-wide energy price. This spatial divergence is the node-level signal GRIDCAST is designed to forecast before the stress event occurs.

A.3 ERA5 and Data Integration

ERA5 (ECMWF global reanalysis, available at cds.climate.copernicus.eu via the `cdsapi` Python library) provides the ground-truth weather input for training at 0.25-degree hourly resolution from 1940 to near-present. All three sources are aligned to UTC before training. Grid Status LMP timestamps vary by ISO and must be converted. Geographic alignment maps EIA Balancing Authority regions to ERA5 grid cells, and Grid Status node coordinates (annotated for approximately 20,000 pricing points) to the nearest ERA5 grid point.